SIMATS ENGINEERING

ASSESSMENT TOOL 3: REVERSE ENGINEERING TASK

Course: Cloud Computing and Big Data Analytics | **Course Code:** CSA1501

Course Outcome: CO4 - Analyze big data characteristics and challenges across domains including security and application aspects.

Student Name	Aakash R	Register No.	192524462
Signature	*Aakash R*	Submission Status	Verified & Completed

Declaration: *I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

1. Big Data Platform for Website, Mobile, and Transaction Data

A probable architecture is a distributed cloud-based data lake and processing platform. Billions of website clicks, mobile interactions, and customer transactions require horizontal scaling, fault tolerance, and partitioned storage.

Storage and ingestion: Raw events can first be collected through an event-streaming layer such as Apache Kafka. The events are then stored in distributed object storage such as Amazon S3 or HDFS. Kafka provides durable, high-throughput ingestion, while the data lake provides inexpensive storage for very large datasets.

Data organization: Structured transaction records can be stored in columnar formats such as Parquet and catalogued in a warehouse or lakehouse. Semi-structured click and mobile-event data can be stored as JSON/Avro records in the data lake. Partitioning by date, region, customer segment, or event type makes queries faster. Metadata catalogs and indexes help analysts locate required datasets.

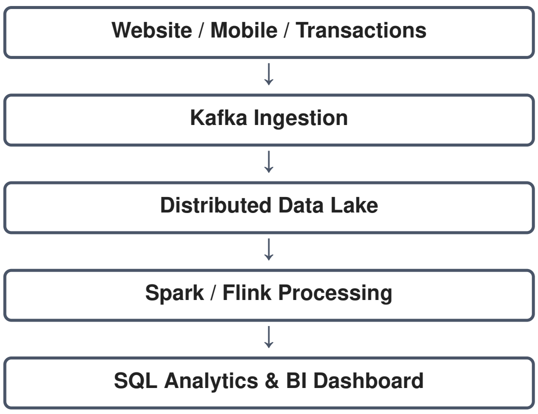
Processing: Apache Spark is a likely batch-processing framework for large-scale transformations, aggregation, ETL, and machine-learning workloads. For real-time analytics, Kafka Streams, Apache Flink, or Spark Structured Streaming could process incoming events with low latency.

Scalability: Data is divided into partitions and processed in parallel across multiple worker nodes. Replication and automatic recovery provide fault tolerance. Partition keys should distribute data evenly and avoid overloaded nodes.

Analytics and reporting: Curated datasets can be exposed through a distributed SQL engine such as Trino/Presto or a cloud data warehouse. Business-intelligence tools can connect to these analytical tables and generate dashboards showing traffic, customer activity, sales, and operational KPIs.

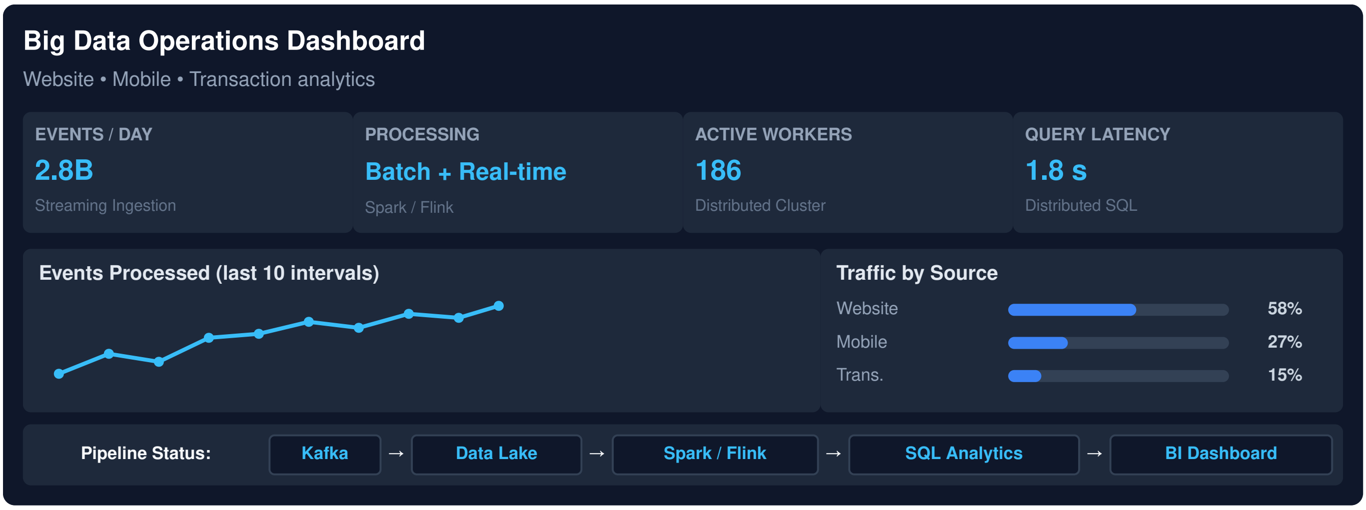
Probable flow: Sources → Kafka → Data Lake → Spark/Flink → Curated Tables → SQL/BI Layer → Dashboards.

Q1 Architecture Flowchart



Representative Output View

Metric	Result
Events	Billions/day
Processing	Batch + Real-time
Analytics	Distributed SQL
Dashboard	Traffic / Sales / Customers



2. E-Commerce Personalization and Recommendation Platform

The platform most likely uses an event-driven architecture because cart additions, product views, abandoned checkouts, and payments arrive continuously from many regions.

Event streaming: Apache Kafka is a probable event backbone. Each event can contain a customer/session ID, product ID, timestamp, region, and event type. Topics can separate product views, carts, payments, and orders.

Customer and session data: A distributed NoSQL database such as Cassandra or DynamoDB could maintain high-volume customer profiles and session information. Redis is suitable for low-latency caching of active sessions, product information, recommendations, and frequently accessed KPIs.

Synchronization: Events from web and mobile applications are written to the streaming layer. Stream-processing tools such as Kafka Streams, Flink, or Spark Structured Streaming can join recent browsing activity with customer profiles and transaction events. This supports real-time personalization.

Analytics and machine learning: Structured order data can be stored in a warehouse/lakehouse, while clickstream JSON can remain in the data lake. Spark or a similar distributed engine can transform and merge both datasets. Machine-learning stages can create features from browsing and purchase history and produce recommendation scores.

Query and dashboards: Trino/Presto or a cloud warehouse can provide distributed analytical queries. BI dashboards can show conversion rate, cart abandonment, revenue, customer activity, recommendation performance, and regional KPIs.

Scalability and fault tolerance: Partition Kafka topics by customer/session or region, replicate messages, and scale consumers horizontally. Replicated storage and checkpointing allow recovery after node failures.

Probable flow: Web/Mobile → Kafka → Stream Processing → Profile/Cache + Data Lake → ML Recommendation → E-Commerce Application; parallel data goes to Warehouse/BI.

Q2 Architecture Flowchart

Web / Mobile Events

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Kafka Ingestion

↓

Stream Processing

↓

Profile DB + Redis

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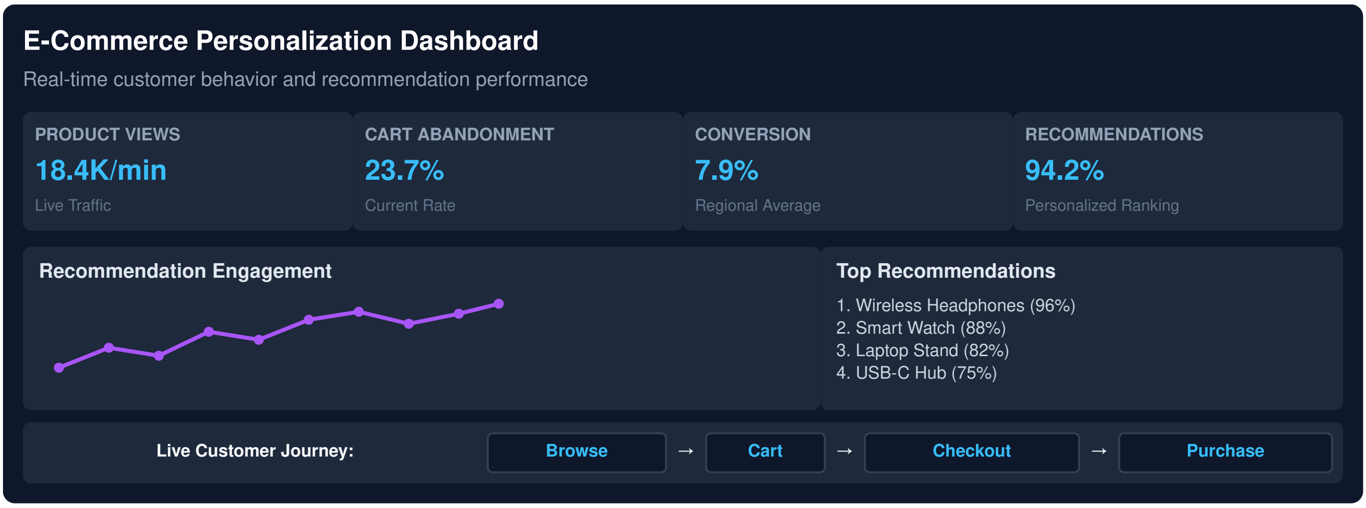
ML Recommendation

Representative Output View

Metric	Result
Product views	Real-time
Cart abandonment	Live KPI
Conversion	Regional trend
Recommendation	Personalized ranking



3. Healthcare Big Data System

A healthcare platform needs a hybrid architecture because patient histories and prescriptions are structured, while diagnostic images and some clinical documents are unstructured or semi-structured.

Storage architecture: A relational database can store structured patient, prescription, laboratory, and billing records. A secure object store or distributed file system can store medical images and documents. A NoSQL database can hold high-volume wearable sensor events and flexible clinical metadata.

Integration: Hospitals, laboratories, and insurance systems can exchange data through interoperability APIs and healthcare standards such as HL7/FHIR. An integration/ETL layer can validate, transform, normalize, and route records into the central platform.

Streaming and analytics: Wearable readings can be ingested through Kafka and processed with Spark Structured Streaming or Apache Flink. Real-time rules can detect abnormal measurements, while batch processing can analyze historical patient populations.

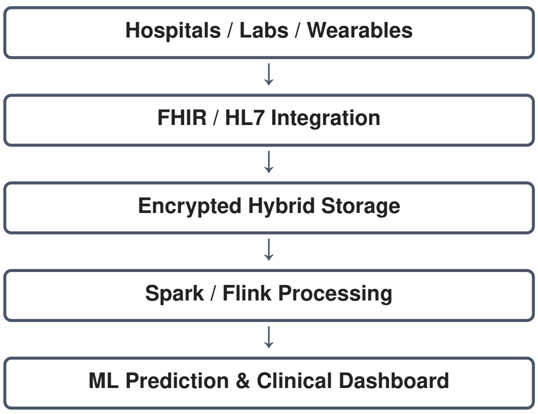
Machine learning: Spark MLlib or another distributed ML platform can train disease-risk and monitoring models. Features may combine patient history, laboratory values, sensor readings, and prescriptions. The resulting risk scores can be presented to authorized clinical users.

Security and compliance: Encryption should protect data at rest and in transit. Role-based or attribute-based access control should restrict sensitive records. Audit logs should record access and changes. Data minimization, retention controls, authentication, and consent management help support healthcare compliance.

Clinical analytics pipeline: Data Sources → Secure Integration → Encrypted Storage → Batch/Stream Processing → ML/Analytics → Authorized Clinical Dashboard.

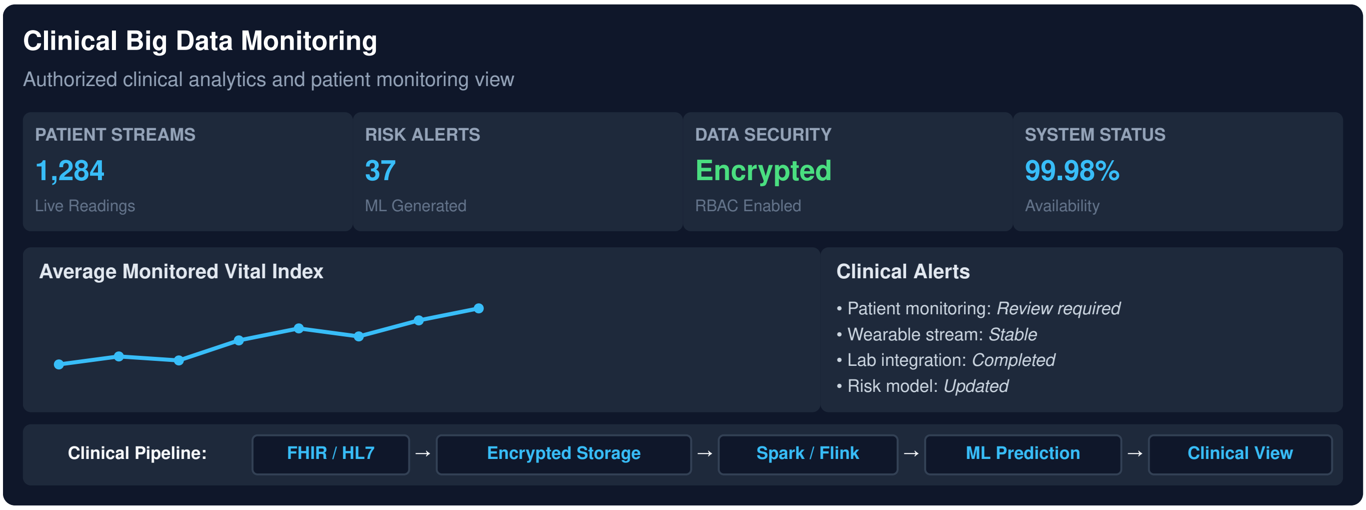
The architecture must prioritize privacy, availability, accuracy, traceability, and controlled access because healthcare data is highly sensitive.

Q3 Architecture Flowchart



Representative Output View

Metric	Result
Patient stream	Live readings
Risk score	ML generated
Security	Encrypted + RBAC
Clinical view	Authorized alerts



4. Smart City Big Data Platform

A smart-city platform benefits from edge computing because traffic sensors, CCTV metadata, weather systems, and public-transport feeds generate continuous data close to where events occur.

Edge layer: Edge gateways can collect and filter sensor data locally. Basic validation, aggregation, and event detection at the edge reduce bandwidth use and latency. Only useful events or summarized data need to be sent to the central cloud.

Messaging and storage: A distributed messaging system such as Kafka can transport continuous sensor events. Centralized object storage can retain historical data. Time-series databases such as InfluxDB or TimescaleDB are suitable for sensor measurements, while a geospatial database such as PostgreSQL with PostGIS can support location-based queries.

Streaming analytics: Flink, Spark Structured Streaming, or Kafka Streams can process traffic and weather events in real time. Geospatial computations can combine vehicle density, road location, weather, and public-transport activity to predict congestion.

Batch plus real-time analytics: Real-time pipelines can support immediate operational decisions, such as congestion alerts. Batch processing can study months or years of traffic patterns for road planning, transport scheduling, and infrastructure investment.

Security: Encryption in transit and at rest, strong authentication, role-based access control, network segmentation, audit logging, and least-privilege permissions can protect citizen and infrastructure data.

Visualization: A BI or operational dashboard can display live traffic conditions, congestion alerts, transport status, weather effects, and historical trends. This gives city operators a unified view of urban conditions.

Probable flow: Sensors/CCTV/Weather/Transport → Edge Gateways → Kafka → Stream Processing → Time-Series/Geospatial Storage → Analytics → Live Dashboard; historical data also enters batch analytics.

Q4 Architecture Flowchart

Sensors / CCTV / Weather / Transport
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Edge Gateway
↓
Kafka Ingestion
↓
Stream + Geospatial Analytics
↓
Time-Series / Geo DB & Live Dashboard

Representative Output View

Metric	Result
Traffic density	Live
Congestion	Predicted
Weather	Correlated
Transport	Live status

Smart City Operations Dashboard

Live traffic, weather and public transport analytics

TRAFFIC DENSITY	CONGESTION	WEATHER EFFECT	TRANSPORT
72%	Predicted	Moderate	94%
City Average	Next 15 min	Correlation Active	Services Active

Live City Traffic Map Status

- **Red:** High congestion zone (Downtown / Ring Road)
- **Amber:** Moderate delays on Airport Expressway
- **Green:** Free flow across suburban corridors

Operational Alerts

- Central Road: *Congestion rising*
- Airport Route: *Normal flow*
- Rain detected: *3 zones*
- Bus network: *94% active*

Edge-to-Cloud Analytics: Sensors → Edge Gateway → Kafka → Stream Analytics → Live Dashboard

5. Healthcare Analytics and Predictive Health System

This system likely combines relational, NoSQL, and object storage because patient records, appointments, wearable readings, and laboratory reports have different data structures.

Hybrid storage: Relational databases can maintain structured patient and appointment records. NoSQL storage can handle high-volume wearable readings. Object storage can preserve reports and other unstructured documents. A data lake can retain raw information before transformation.

ETL and interoperability: ETL pipelines can extract data from hospitals and clinics, validate it, standardize fields, remove duplicates, and load curated datasets. HL7/FHIR-based interfaces can support interoperability between healthcare applications.

Real-time monitoring: Wearable readings can flow through Kafka and be processed using Flink or Spark Structured Streaming. Threshold rules and streaming models can identify potentially abnormal patterns and generate alerts for authorized users.

Population-level processing: Apache Spark can process large historical datasets in parallel. Batch jobs can calculate disease trends, treatment outcomes, risk factors, and population statistics.

Security and governance: Encryption, role-based access control, strong authentication, audit trails, data retention policies, and compliance controls should be applied throughout the pipeline. Sensitive data should only be available to authorized personnel.

Machine learning: Models can combine demographics, medical history, laboratory results, wearable features, and appointment patterns to produce risk scores. A clinical decision-support layer can present model outputs as recommendations or alerts, while final decisions remain with qualified healthcare professionals.

Probable flow: Hospital/Clinic + Wearables + Labs → Secure ETL/Interoperability → Hybrid Storage → Spark/Flink → ML Models → Risk Scores/Analytics → Authorized Dashboard.

Q5 Architecture Flowchart

Patient / Wearable / Lab Data

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Secure ETL

↓

Hybrid Storage

↓

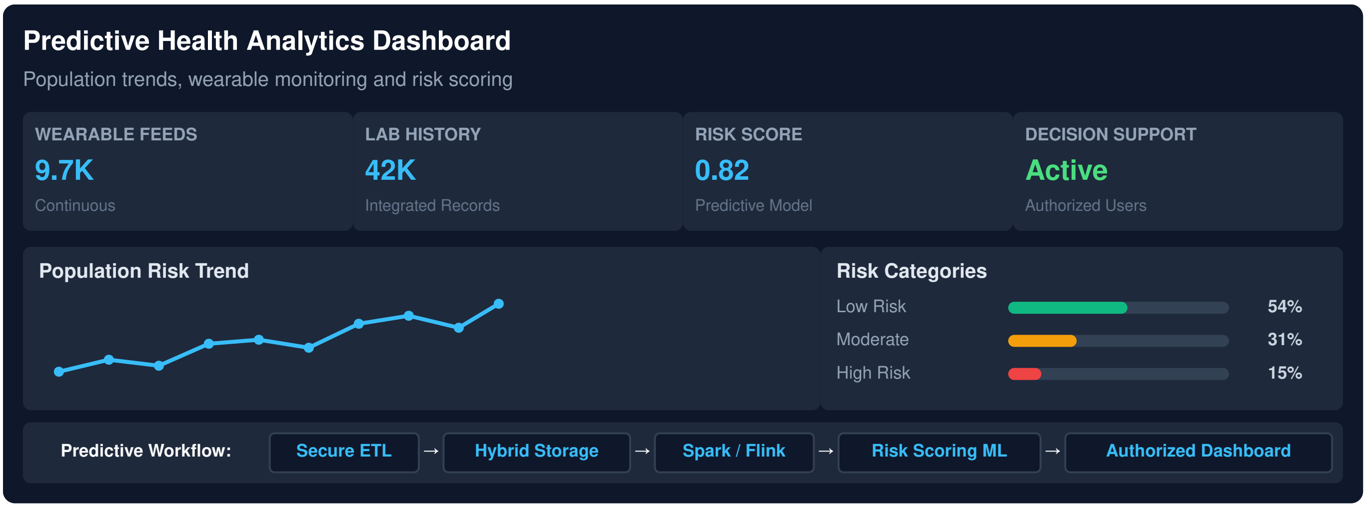
Spark / Flink

↓

Risk Scoring ML & Authorized Dashboard

Representative Output View

Metric	Result
Wearables	Continuous
Lab history	Integrated
Risk score	Predictive
Decision support	Authorized



Architecture Summary

Across the five cases, the recurring reverse-engineered pattern is: **data sources → ingestion/message streaming → distributed storage → batch and stream processing → machine learning/analytics → SQL/BI/operational dashboards**. Partitioning, replication, horizontal scaling, caching, encryption, access control, and audit logging are key design decisions for reliable big-data platforms.

Assessment Coverage

The assessment contains 5 questions for 30 marks and emphasizes inference accuracy, architecture understanding, technical evidence, depth of analysis, and clarity.

Flowchart Diagrams and Output Views Summary

The following flowcharts show the probable architecture inferred for each assessment task. The output views are illustrative representations, not fabricated software screenshots.

Question 1 - Big Data Platform

Website / Mobile / Transactions → Kafka Ingestion → Distributed Data Lake → Spark/Flink → SQL Analytics → BI Dashboard

Question 2 - E-Commerce Personalization

Web / Mobile Events → Kafka → Stream Processing → Profile DB + Redis → ML Recommendation → KPI Dashboard

Question 3 - Healthcare Big Data

Hospitals / Labs / Wearables → FHIR/HL7 Integration → Encrypted Hybrid Storage → Spark/Flink → ML Prediction → Clinical Dashboard

Question 4 - Smart City

Sensors / CCTV / Weather / Transport → Edge Gateway → Kafka → Stream + Geospatial Analytics → Time-Series/Geo DB → Live Dashboard

Question 5 - Healthcare Analytics

Patient / Wearable / Lab Data → Secure ETL → Hybrid Storage → Spark/Flink → Risk Scoring ML → Authorized Dashboard

Representative Output Screens

These compact output tables represent the type of result that would appear on an analytics or monitoring dashboard.

1. Big Data Platform

Output	Representative Result
Events	Billions/day
Processing	Batch + Real-time
Analytics	Distributed SQL
Dashboard	Traffic / Sales / Customers

2. E-Commerce

Output	Representative Result
Product views	Real-time
Cart abandonment	Live KPI
Conversion	Regional trend
Recommendation	Personalized ranking

3. Healthcare

Output	Representative Result
Patient stream	Live readings
Risk score	ML generated
Security	Encrypted + RBAC
Clinical view	Authorized alerts

4. Smart City

Output	Representative Result
Traffic density	Live
Congestion	Predicted
Weather	Correlated
Transport	Live status

5. Healthcare Analytics

Output	Representative Result
Wearables	Continuous
Lab history	Integrated
Risk score	Predictive
Decision support	Authorized